

$$\begin{aligned}
 (ii) \quad (256)^{0.16} \times (16)^{0.18} &= \{(16)^2\}^{0.16} \times (16)^{0.18} = (16)^{(2 \times 0.16)} \times (16)^{0.18} \\
 &= (16)^{0.32} \times (16)^{0.18} = (16)^{(0.32+0.18)} = (16)^{0.5} = (16)^{\frac{1}{2}} = 4.
 \end{aligned}$$

Ex. 4. Solve : $9^{8.6} \times 8^{3.9} \times 72^{4.4} \times 9^{3.9} \times 8^{8.6} = 72^x$.

(L.I.C., 2005)

Sol. Let $9^{8.6} \times 8^{3.9} \times 72^{4.4} \times 9^{3.9} \times 8^{8.6} = 72^x$.

$$\text{Then, } 9^{(8.6+3.9)} \times 8^{(3.9+8.6)} \times 72^{4.4} = 72^x$$

$$\Leftrightarrow 9^{12.5} \times 8^{12.5} \times 72^{4.4} = 72^x \Leftrightarrow (9 \times 8)^{12.5} \times 72^{4.4} = 72^x$$

$$\Leftrightarrow 72^{12.5} \times 72^{4.4} = 72^x \Leftrightarrow 72^{(12.5+4.4)} = 72^x \Leftrightarrow 72^{16.9} = 72^x \Leftrightarrow x = 16.9.$$

Ex. 5. Solve : $(0.064) \times (0.4)^7 = (0.4)^x \times (0.0256)^2$.

(Bank P.O., 2010)

Sol. Let $(0.064) \times (0.4)^7 = (0.4)^x \times (0.0256)^2$.

$$\text{Then, } (0.4)^3 \times (0.4)^7 = (0.4)^x \times [(0.4)^4]^2$$

$$\Leftrightarrow (0.4)^{(3+7)} = (0.4)^x \times (0.4)^8 \Leftrightarrow (0.4)^{10} = (0.4)^{x+8} \Leftrightarrow x+8=10 \Leftrightarrow x=2.$$

Ex. 6. What is the quotient when $(x^{-1} - 1)$ is divided by $(x - 1)$?

Sol.
$$\frac{x^{-1} - 1}{x - 1} = \frac{\frac{1}{x} - 1}{x - 1} = \frac{(1 - x)}{x} \times \frac{1}{(x - 1)} = -\frac{1}{x}.$$

Hence, the required quotient is $-\frac{1}{x}$.

Ex. 7. If $2^{x-1} + 2^{x+1} = 1280$, then find the value of x .

Sol. $2^{x-1} + 2^{x+1} = 1280 \Leftrightarrow 2^{x-1} (1 + 2^2) = 1280$

$$\Leftrightarrow 2^{x-1} = \frac{1280}{5} = 256 = 2^8 \Leftrightarrow x-1=8 \Leftrightarrow x=9.$$

Hence, $x = 9$.

Ex. 8. Find the value of $\left[5 \left(8^{\frac{1}{3}} + 27^{\frac{1}{3}}\right)^3\right]^{\frac{1}{4}}$.

Sol.
$$\begin{aligned} \left[5 \left(8^{\frac{1}{3}} + 27^{\frac{1}{3}}\right)^3\right]^{\frac{1}{4}} &= \left[5 \left\{(2^3)^{\frac{1}{3}} + (3^3)^{\frac{1}{3}}\right\}^3\right]^{\frac{1}{4}} = \left[5 \left\{2^{\left(3 \times \frac{1}{3}\right)} + 3^{\left(3 \times \frac{1}{3}\right)}\right\}^3\right]^{\frac{1}{4}} \\ &= \{5(2+3)^3\}^{\frac{1}{4}} = (5 \times 5^3)^{\frac{1}{4}} = (5^4)^{\frac{1}{4}} = 5^{\left(\frac{4 \times 1}{4}\right)} = 5^1 = 5. \end{aligned}$$

Ex. 9. Find the value of $\left\{(16)^{\frac{3}{2}} + (16)^{-\frac{3}{2}}\right\}$.

Sol.
$$\begin{aligned} \left[(16)^{\frac{3}{2}} + (16)^{-\frac{3}{2}}\right] &= \left[(4^2)^{\frac{3}{2}} + (4^2)^{-\frac{3}{2}}\right] = 4^{\left(2 \times \frac{3}{2}\right)} + 4^{\left\{2 \times \left(\frac{-3}{2}\right)\right\}} \\ &= 4^3 + 4^{-3} = 4^3 + \frac{1}{4^3} = \left(64 + \frac{1}{64}\right) = \frac{4097}{64}. \end{aligned}$$

Ex. 10. If $\left(\frac{1}{5}\right)^{3y} = 0.008$, then find the value of $(0.25)^y$.

Sol.
$$\left(\frac{1}{5}\right)^{3y} = 0.008 = \frac{8}{1000} = \frac{1}{125} = \left(\frac{1}{5}\right)^3 \Leftrightarrow 3y = 3 \Leftrightarrow y = 1.$$

$$\therefore (0.25)^y = (0.25)^1 = 0.25.$$